SMART CLEANING ROBOT

ME-GY 5103 MECHATRONICS

Group-4

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Abstract

To achieve intelligent cleaning, research must traditionally be carried out in areas such as sensor technology and path planning technology. Due to time constraints, this project only used sensor technology to simulate the normal cleaning process of an intelligent cleaning robot. This report shows the development process of this project step by step, from project establishment, analysis of the characteristics of best-selling products on the market, and the design of this project.

This report contains information such as theoretical basis, background analysis, bill of materials, mechanical design and electronic circuit. It also describes in detail the reasons for using each component. In addition, after the product has been successfully operated, an analysis of the advantages and disadvantages has been carried out, and the cost of the prototype and mass production has been considered.

1. Introduction

1.1. Executive summary

The production of intelligent cleaning robots predominantly use injection molding for shell fabrication [1]. Subsequently, components are integrated into the cleaning robot chassis on the assembly line to facilitate mass production. This project modifies the method by adding laser cutting, which employs a more individualized approach. This project is a basic replica of the top-selling robotic vacuum cleaner available in the market.
The project initially enumerated the existing functions attainable through the analysis of intelligent cleaning robots' characteristics and the integration of acquired knowledge. This aids in determining which attributes should be contemplated in the design. Subsequently, performance benchmarks were established to ensure the project's product attained a comparably efficient operational velocity. The circuit connections and PBasic code were then devised to guarantee the appropriate functioning of all components. The shell was ultimately reengineered to incorporate acrylic, enhancing its visual appeal, while simultaneously rendering the interior structure transparent and reinforcing its durability. This project utilizes CAD drawing software to present three views of the product along with detailed specifications for each component. The project fabricated the enclosure utilizing laser cutting and 3D printing technology. The cumulative duration for the production and assembly procedure is around 8 hours. Excluding the cost of the production machinery, the total expense for all raw materials amounts to — USD. If the product is deemed suitable for mass production, assembly line integration is necessary, and the cost of the associated machinery and labor is — USD.

1.2. Background Check

The advancement of computer and electronic technology has rendered smart homes a focal point of interest. A significant quantity of cleaning robots has emerged in the market, with intelligent cleaning robots being a primary category of smart home gadgets [2]. The future

trajectory of cleaning robots is toward increased intelligence and convenience [3]. Since the 1880s, vacuum cleaner designs have progressively transitioned from conventional horizontal and vertical models to advanced robotic cleaning devices [4]. Since 2001, when the inaugural mass-produced intelligent robot was created by the Swedish home appliance company Leks, developed nations in Europe [5], America, and Japan have subsequently introduced functional intelligent cleaning robots to the market. By 2024, the global annual average export volume of items will surpass 28 million units, with revenue in this sector attaining an annual average of 5 billion U.S. dollars [6].
This project has researched the development and deployment of intelligent cleaning robots. The primary research focus is on the investigation of its essential sensor technology. The PBasic program renders the cleaning robot both straightforward and intelligent. By examining the casings of the top-selling intelligent cleaning robots in recent years, the shell appropriate for this product has been reengineered, with the intention of anticipating the future prospects and trends in the intelligent cleaning robot sector.

Figure I :Appearance of current best-selling intelligent cleaning robot

2. Mechanical design

2.1. Characterization

ROBOT NAVIGATION:

- Four DC motors control the robot's movement in all directions.
- An ultrasonic sensor at the front detects obstacles and guides navigation.

CLEANING MECHANISM:

- DRY CLEANING MODE: A front servo motor operates a dry cleaning brush for sweeping.
- WET CLEANING MODE: A rear servo motor controls a mopping brush for wet cleaning.

WATER SUPPLY SYSTEM:

- A soil moisture sensor in the water tank monitors water levels.
- A water pump sprays water onto the ground for mopping when water is sufficient.

FEEDBACK AND CONTROL LOOP:

- If the water tank is empty, the LCD displays a warning and prompts the user to refill.
- The bot automatically switches to dry cleaning mode, stopping the rear servo motor.

DISPLAY AND USER INTERACTION:

- The LCD shows "Tank Full" or "Tank Empty" messages to inform the user of the water tank status.
- Real-time operational updates are displayed for better user understanding.

POWER CONTROL AND SAFETY:

- An external tilt switch is provided for safe ON/OFF operation.
- Feedback systems ensure safe transitions between cleaning modes and prevent damage to components.

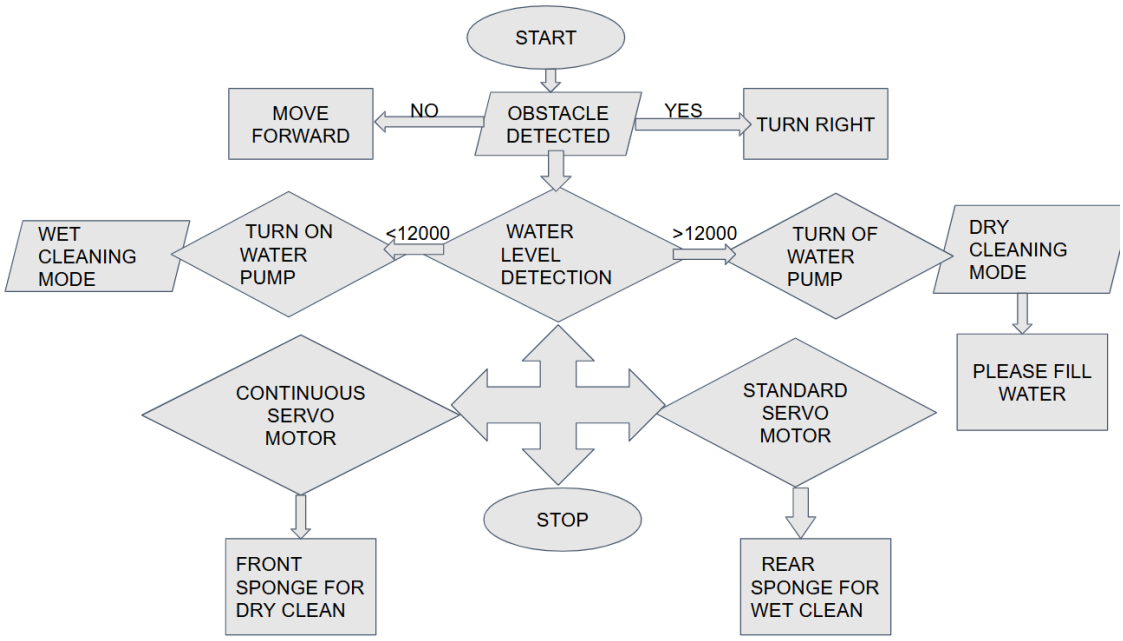

Figure II :Workflow of intelligent cleaning robot

2.2. Bill-of-material

The bill-of-material is shown below in the table:

Equipment name	Quantity	Cost
Parallax Basic Stamp2	1	$235
Water Pump	1	$6
Ultrasonic Sensor	1	$5
Motor Driver	2	$6
DC Motor	4	$5
Motor Wheel	4	$4
Servo Motor	2	$8
Li-Bo Battery	1	$10
Moisture Sensor	1	$2
Resister (10k)	1	$1
Capacitor (0.1uf)	1	$1
Water Tank (3D Print)	1	$2
Cleaning Brush	2	$2
Jumper Wires	Several	$5
Acrylic Shell	Several parts	$15
TOTAL		$307

Table 1 ：Bill-of-material

The following sections will select important components and explain in detail their role and the reasons for using them.

2.2.1. Ultrasonic sensor

The ultrasonic sensor operates on the idea of detecting barriers by emitting and receiving ultrasonic waves. Upon receiving the reflected ultrasonic signal, the intelligent cleaning robot can ascertain the distance to the barrier, so preventing a collision [7]. In intricate surroundings, ultrasonic sensors can facilitate navigation and, when integrated with additional sensors (such as lidar and infrared sensors), yield more precise positioning data.

This project involves the installation of one ultrasonic sensors on the front sides of the cleaning robot. The operational logic of ultrasonic sensor is as follows:

1. The safety distance for the ultrasonic sensor positioned directly in front of the robot is 50 cm. When the safety distance is 50 cm or less, the further step will be undertaken to ascertain the next course of action.
2. Upon evaluating judgment condition 1, if the safety distance exceeds 50 cm, the robot will move forward. If the safety distance is 50 cm or less, the robot will execute to do right right turns to perform a right U-turn. Execute this technique in an enclosed space, and upon completion of several iterations, we get to sanitize the entire area.

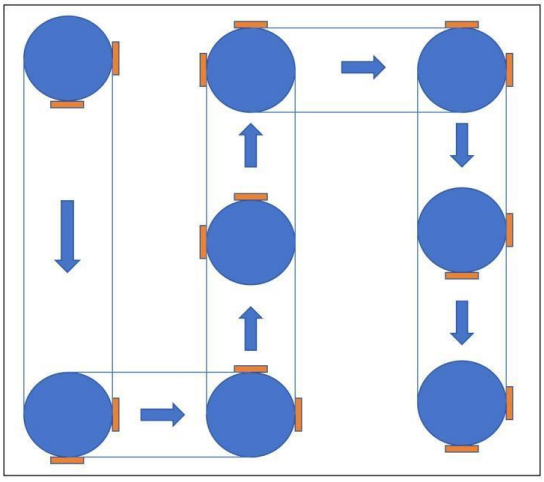

Figure III :Diagram of the cleaning process

2.2.2. Wheel, DC motors and motor drivers

The wheels, DC motor, and motor driver collaborate to constitute the robot's locomotion components.
The wheels serve a straightforward function in the intelligent cleaning robot, however the DC motor and motor driver each fulfill a significant role:

The DC motor serves as the primary power source for the intelligent cleaning robot, facilitating the movement of its wheels or tracks. The velocity is regulated by modifying the voltage and current via a predetermined remote control program, thus attaining various cleaning speeds and adaptable movement. The DC motor exhibits substantial starting torque and rapid response speed, enabling swift initiation and cessation to accommodate diverse requirements during the cleaning procedure.

The motor driver governs the operational condition of the DC motor, encompassing initiation, cessation, forward rotation, and reverse rotation. This project employs the PWM approach to operate the motor. Consequently, the motor driver can modulate the motor's voltage via pulse width modulation (PWM) and other methods to get accurate regulation of motor speed and torque. The motor driver typically includes overcurrent, overvoltage, and overheating protection mechanisms to safeguard both the motor and the driver, hence ensuring the system's safety and reliability. The circuit created for this project enables the driver to interface with the Basic Stamp microcontroller and receive signals from the ADC. Each driver orchestrates the operation of two motors to generate controllable motion patterns.

Consequently, in the project's architecture, the DC motor supplies power and motion regulation, whereas the motor driver oversees the motor's operational state and performance. The collaboration of the two guarantees that the robot functions efficiently, adaptively, and intelligently.

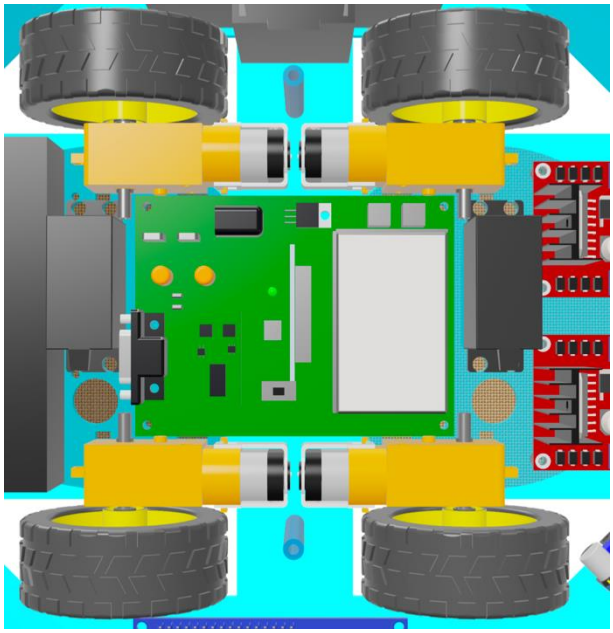

Figure IV :DC motor and motor drive

2.2.3. Humidity sensor, servo motor, pump and water tank

The humidity sensor, servo motor, and water tank collectively constitute the cleaning mechanism of the product.

The product is designed to utilize a damp sponge for cleaning purposes. The sponge is linked to a servo motor. The servo motor offers precise angular control, enabling the sponge to rotate at a specified angle. A Basic Stamp2 microprocessor drives the servo motor. Similar to a DC motor, the rotational speed and direction of the sponge can be changed by a pre-programmed remote control. This enhances the cleaning process's flexibility and intelligence. Furthermore, the servo motor can deliver adequate torque to rotate the sponge and sustain stable rotation while cleaning a soiled floor.

The primary role of the humidity sensor is to measure the humidity level of the cleaning environment in real time, assisting the robot in assessing the wetness of the region. The robot may autonomously modify the cleaning technique based on humidity sensor readings. In regions with elevated humidity, the robot may use the wet mopping mode, whereas in arid locations, it might employ the dry sweeping mode. The humidity sensor is integrated inside the cleaning sponge in the product's design. The humidity sensor's job is

to measure the sponge's moisture levels in real time as the wet sponge progressively loses moisture through rotation and friction with the ground during the cleaning process. When the sponge's moisture level is insufficient, a DC motor will activate to transfer water from the tank to the sponge via the water pipe, thereby maintaining the sponge's moisture and aiding the subsequent cleaning procedure.

2.3. Circuit flow and Connection

Combining the above sections, the analog circuit flow and actual circuit connections for this product are shown below.

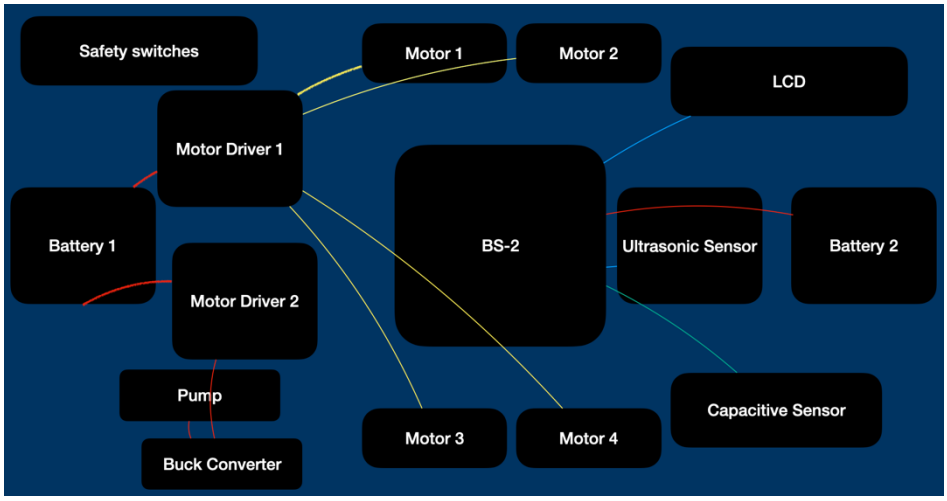

Figure VIII :Connection of circuit

2.4. BASIC STAMP -2 CODE

```
' {$STAMP BS2}
' {$PBASIC 2.5}

' Constants and Variables
CmConstant CON 2260       ' Conversion constant for distance in cm
WATER      CON 9          ' Water pump is connected to pin 9
cmDistance VAR Word       ' Variable to store distance in cm
time       VAR Word       ' Variable to store echo pulse time
TransTime  VAR Word       ' Variable to store RCTime for Ultrasonic sensor
counter1 VAR Byte         'FOR SERVO MOTOR 1
counter2 VAR Byte         'FOR SERVO MOTOR 2

' Motor Control Pins
SideA_Fwd   PIN 14     ' IN14 for Channel A (Side A forward)
SideA_Rev   PIN 13     ' IN13 for Channel A (Side A reverse)
SideB_Fwd   PIN 12     ' IN12 for Channel B (Side B forward)
SideB_Rev   PIN 11     ' IN11 for Channel B (Side B reverse)
MOTORA      PIN 6      ' Water Pump pin 6
MOTORB      PIN 7      ' Water Pump pin 7

'LCD Display
SEROUT 0,84,[22,12]    'CLEAR ALL

'MAIN CODE
Main:

  'CAPACITIVE SOIL MOISTURE CODE:
  HIGH WATER
  PAUSE 5
  RCTIME WATER, 1, TransTime
  TransTime= Transtime *2
  DEBUG DEC Transtime ,13
  IF Transtime >12000 THEN
    LOW MOTORA          ' Turn on the motor
    LOW MOTORB
```

```
  DEBUG "DRY" ,13
  SEROUT 0,84,["DRY CLEANING MODE : ",13,"WATER LEVEL LOW -> PLEASE FILL
WATER" ]

 ELSE
  HIGH MOTORA          ' Turn off the motor
  LOW MOTORB
  DEBUG "WET",13
  SEROUT 0,84,["WET CLEANING MODE : ",13,"TANK FULL"]
 ENDIF

 'STANDARD SERVO MOTOR CODE FOR MOPPING:
 FOR  counter1=1 TO 50
   PULSOUT 15, 250      '0 Degree
   PAUSE 1
 NEXT
 FOR  counter1=1 TO 50
   PULSOUT 15, 1250     '180 Degree
   PAUSE 1
 NEXT


 'CONTINUOUS SERVO MOTOR CODE FOR SWEEPING:
 PULSOUT 8, 1000      '0

 'ULTRASONIC SENSOR WITH 4 DC MOTORS FOR MOTION

 PULSOUT 10, 5                        ' Send trigger pulse pin 15
 PULSIN 10,1, time                    ' Measure echo pulse time
 cmDistance = CmConstant ** time   ' Convert time to distance in cm

                                     ' Decision-making based on distance
 IF cmDistance <= 50 THEN          ' If object detected within 50 cm
  GOSUB TurnRight                    ' Turn the robot to the right
 ELSE
  GOSUB MoveForward                  ' Else move the robot forward
 ENDIF
```

```
  PAUSE 100                          ' Small delay before next loop

GOTO Main                            ' Repeat the process


MoveForward:                          ' Subroutine to move forward
  HIGH SideA_Fwd                       ' Side A forward
  LOW SideA_Rev
  HIGH SideB_Fwd                      ' Side B forward
  LOW SideB_Rev
   DEBUG "Moving Forward" ,13
  RETURN


TurnRight:                           ' Subroutine to turn right
  HIGH SideA_Fwd                    ' Side A forward
  LOW SideA_Rev
  LOW SideB_Fwd                     ' Side B stop
  HIGH SideB_Rev                   ' Side B reverse
  DEBUG "Turn Right" ,13
  RETURN
```

2.5. Shell design and manufacturing

The product's exterior was reconfigured to resemble the shape of the top-selling intelligent cleaning robot, with modifications made to accommodate its electrical connections. The outcome is a visually appealing product that permits clear visibility of its internal features. The components of the outer shell were fabricated from a 24 ft × 12 ft acrylic sheet utilizing laser-cutting technology. Screw posts are produced with 3D printing technology to secure essential components, such as the Basic Stamp, to the shell. The DC motor and other components are affixed to the base of the shell with robust adhesive tape. The resultant shell possesses the benefits of strength, lightness, and customizability.

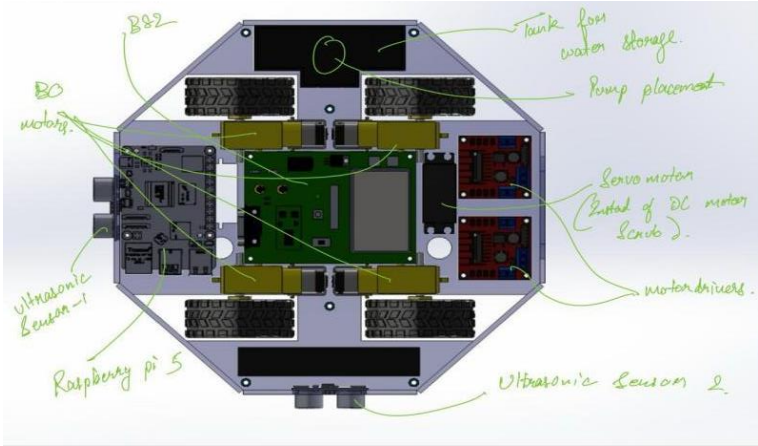

Figure IX :Location of components

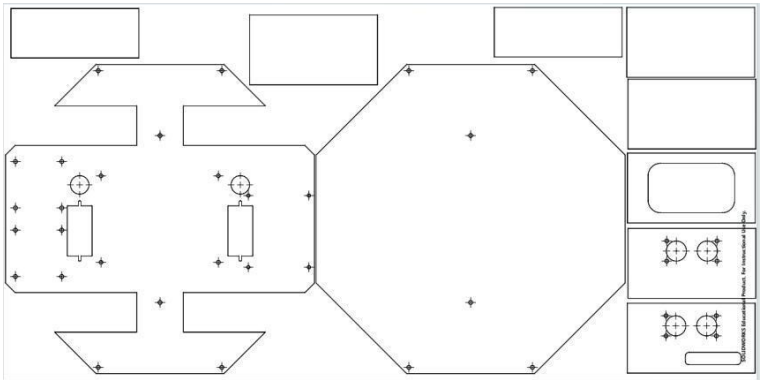

Figure X :CAD to be laser-cutting

3. Results and conclusion

3.1. Assembly and actual testing

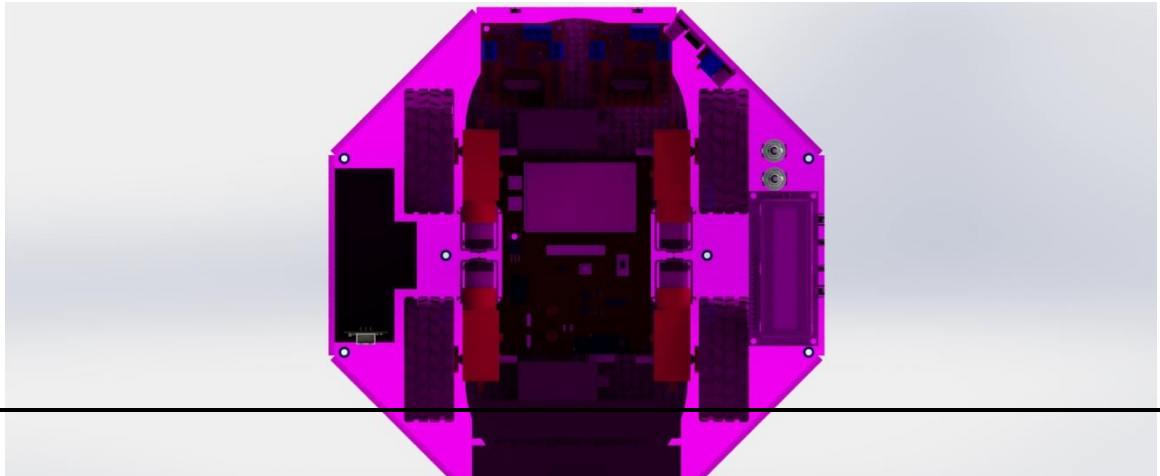

Figure XI :Appearance of the robot after assembly

Figure XII :The actual testing process

3.2. Results

As can be seen in the demonstration video, the smart cleaning robot integrates the technology to deliver efficient and autonomous cleaning performance, addressing both dry and wet

cleaning needs. Equipped with dual cleaning modes, the front-mounted brush powered by a servo motor handles dry cleaning tasks, while the rear-mounted brush performs wet mopping, enhanced by an automated water spraying system. A soil moisture sensor monitors water levels in the tank, seamlessly switching to dry cleaning mode when the tank is empty. This ensures uninterrupted operation and extends the robot's utility without manual intervention. Additionally, an ultrasonic sensor enables obstacle detection, ensuring smooth navigation and collision avoidance for hassle-free operation.

To enhance user interaction and safety, the robot features an LCD display that provides real-time updates on the water tank status, alerting users when refilling is required. The external tilt switch offers safe and easy ON/OFF control, while feedback-driven systems protect motors from damage due to low water levels. Designed for autonomy, the robot minimizes manual effort by automatically adjusting between cleaning modes based on sensor inputs. Its versatile functionality, coupled with user-friendly feedback systems, makes it an ideal solution for household and office cleaning, merging convenience with cutting-edge technology.

3.3. Reflection of advantages and disadvantages

Advantages

1. It operates and cleans independently, minimizing the time and effort required for manual cleaning and enhancing convenience.
2. It can proficiently detect and circumvent obstructions promptly using ultrasonic sensors and algorithms, thereby enhancing cleaning efficiency.
3. It can dispense water to the cleaning sponge based on the level of saturation to maintain its moisture.
4. The robot's velocity and trajectory can be remotely managed using a smartphone application.
5. It features a compact and customized design, enabling access to confined areas for cleaning and enhancing overall cleaning coverage.

6. It is also comparatively tranquil throughout operation, ensuring it does not disrupt your daily activities.

Disadvantages

1. Notwithstanding its uncomplicated form, the robot's total expense is elevated owing to components including materials, sensors, and motors. An unoptimized design may lead to material waste or intricate constructions.
2. It is ineffective at removing entrenched dirt or navigating intricate situations.
3. The restricted battery capacity results in limited cleaning duration, necessitating frequent recharging, hence diminishing user convenience.

3.4. Conclusion

This project entails the design and execution of a basic intelligent cleaning robot, intended to enhance the productivity and ease of domestic cleaning via automated cleaning technologies. The robot integrates multiple sensors and motor control systems to efficiently clean the floor. The robot effectively executes autonomous driving functions, identifies obstacles, and circumvents them to guarantee safe operation. The cleaning efficacy is commendable, as it successfully eliminates dust and debris from the floor. Lightweight acrylic shell material was utilized in the design process. The motor and sensors are strategically located to guarantee the robot's movement and stability. The trajectory and velocity of movement can be effortlessly modified using an intuitive user interface, enhancing usability. The robot functions silently, which is essential for residential applications.

Nonetheless, the initiative encountered several challenges. The elevated cost of the components utilized in the experiment precluded the mass production of the identical configuration. The basic cleaning apparatus yielded suboptimal cleaning efficiency. The longevity of the battery has also emerged as a problem. While the robot is capable of doing a singular cleaning duty, its battery life constrains its operational duration when addressing allareas.

This research illustrated the design and deployment of a basic intelligent cleaning robot and validated its use in domestic cleaning. Future endeavors may concentrate on

augmenting the robot's intelligence, increasing battery longevity, and enhancing environmental adaptation through the implementation of technologies such as pattern recognition and path planning. This will augment its functionality and user experience. This research has enhanced the team's comprehension of intelligent robotics technology and established a foundation for the design of more intricate robots in the future. Future research is anticipated to integrate additional sensors and sophisticated algorithms to create more efficient and intelligent cleaning solutions.

4. Manufacturing strategies

4.1. Prototype cost

Components name	Quantity	Price
Parallax Basic Stamp2	1	$235
Water Pump	1	$6
Ultrasonic Sensor	1	$5
Motor Driver	2	$6
DC Motor	4	$5
Motor Wheel	4	$4
Servo Motor	2	$8
Li-Bo Battery	1	$10
Moisture Sensor	1	$2
Resister (10k)	1	$1
Capacitor (0.1uf)	1	$1
Water Tank (3D Print)	1	$2
Cleaning Brush	2	$2
Jumper Wires	Several	$5
Acrylic Shell	Several parts	$15
TOTAL		$307

Table 2 ： Total cost of components

Regardless of the time spent on market research, regardless of the time and effort spent on designing everything from the circuit connections to the housing, and regardless of the cost of 3D printers and laser-cutting machines. Only the cost of purchasing the individual components is considered (from electronics stores or online shopping such as Amazon), and the cost of each component is shown in the chart.

4.2. Cost analysis for mass production

If the intelligent cleaning robot designed in this project is put into mass production, the difference between the mass production cost and the prototype cost needs to be analyzed first. By reading relevant papers, it is known that the difference is mainly reflected in the following aspects:

1. Material cost

When manufacturing a prototype, small batches or single pieces of material are usually used, and the price will be higher due to the small purchase quantity. If the product is put into mass production, bulk discounts can be enjoyed by purchasing materials on a large scale, which will reduce the unit material cost.

2. Production process

For the manufacturing process of this project, all parts are completed manually by the team members (3 people). If the preliminary investigation and design stages are not included. It is estimated that the manufacturing process of the shell, connecting the circuit and assembly takes about 8 hours. This production model is inefficient, with high labor and time costs. If the product enters mass production, an automated production line can be used to improve production efficiency and reduce the production time and labor costs per product. The difficulty lies in the installation price of the automated production line and subsequent maintenance.

3. Equipment investment

The equipment used for assembly in this project is standardized and consists of a set of screwdrivers, a bottle of self-adhesive tape and a set of soldering iron tools. The composition of the tools is very basic. If the product is to be mass-produced, investment in special equipment and injection molds will be required, which will increase the initial investment cost. The cost difference for this is estimated at 150 USD.

4. Design optimization

As discussed in the previous section, the design of this project may have problems with material waste or complex structures. If the product is to be mass-produced, additional efforts will be required to optimize it, so as to reduce material usage, simplify the assembly process and improve manufacturability. The estimated cost difference for this is 20 USD.

5. Testing and verification

Although the actual tests in the previous section were successful, the number of tests was small, and design flaws may not have been discovered. Examples include component wear, battery life, and circuit connections. If the product is to be mass-produced, rigorous quality control and testing will be required to ensure that each product meets the standards. The cost difference for this is estimated at USD.

Therefore, the final estimated cost of mass production of 100 such smart cleaning robot is is 28,000 USD by considering 280 USD per robot.

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